Chebotarev density theorem

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Abstract

This document outlines a Lean 4 / Mathlib formalisation of the **Chebotarev density theorem** in conjugacy-class form. For a finite Galois extension L/K of number fields with Galois group $G = \mathrm{Gal}(L/K)$ and a conjugacy class $C \subseteq G$, the Dirichlet density of the set of primes $\mathfrak{p}$ of $\mathcal{O}_K$ whose Frobenius conjugacy class is C equals $|C| \,/\, |G|$.

The proof follows the modern exposition of Stevenhagen–Lenstra (*Chebotarëv and his density theorem*, Appendix) and Sharifi (*Algebraic Number Theory*, §7.2). Both references are included under `docs/`.

Outline. The proof proceeds in three reductions:

1. *Foundations* (Chapters 1 and 2). We define the Dirichlet density of a set of prime ideals via the Dirichlet series $\sum_{\mathfrak{p}\in S} \mathrm{N}(\mathfrak{p})^{-s}$ normalised against the same sum over all primes, and we set up the Frobenius element at an unramified prime $\mathfrak{P}$ of $\mathcal{O}_L$ together with its conjugacy class as $\mathfrak{P}$ varies above a K-prime $\mathfrak{p}$.

2. *Cyclotomic case* (Chapters 3 and 4). The analytic engine is the factorisation $\zeta_L(s) = \prod_{\chi\in\widehat{G}} L(\chi, s)$ of the Dedekind zeta function of L into Hecke (Artin) L-functions indexed by characters of G, valid when G is abelian, together with the non-vanishing $L(\chi, 1) \neq 0$ for nontrivial χ. Applied to a cyclotomic extension $L = K(\zeta_m)$, this yields the cyclotomic case of Chebotarev directly from Dirichlet's argument.

3. *Abelian case* (Chapter 5). For a general abelian L/K, we use Chebotarev's crossing trick: pick m coprime to the discriminant of L so that $\mathrm{Gal}(L(\zeta_m)/K) \cong G \times H$ with $H = \mathrm{Gal}(K(\zeta_m)/K)$; for $\tau \in H$ with $|G| \mid \mathrm{ord}(\tau)$, the cyclic subgroup $\langle(\sigma, \tau)\rangle \subseteq G \times H$ has trivial intersection with $G \times \{1\}$, so the fixed-field extension is cyclotomic; the cyclotomic case applies; summing over admissible τ and letting $m \equiv 1 \pmod{n^k}$ for $k \to \infty$ recovers the abelian density.

4. *General case* (Chapter 6). The conjugacy-class form is reduced to the cyclic-extension case via the intermediate field $E = L^{\langle\sigma\rangle}$: the counting "exactly $|G| \,/(f\,|C|)$ primes of L above $\mathfrak{p}$ have Frobenius σ" yields the density relation $\delta_K(S) = (f\,|C| \,/\, |G|)\delta_E(T_\sigma)$, which combines with the abelian case for L/E.

The proof needs no class field theory; it is essentially Chebotarev's original strategy as expounded by Stevenhagen–Lenstra.

Contents

Chapter 1

Dirichlet density

1.1 Dirichlet density of a set of prime ideals

Throughout, K denotes a number field, $\mathcal{O}_K$ its ring of integers, and $\mathfrak{p}$ ranges over nonzero prime ideals of $\mathcal{O}_K$ with absolute norm $N\mathfrak{p} = [\mathcal{O}_K : \mathfrak{p}]$.

Definition 1.1.1. *A set S of nonzero prime ideals of $\mathcal{O}_K$ has* Dirichlet density $\delta \in \mathbb{R}$*, written $\delta(S) = \delta$, if*

$$\lim_{s \to 1^+} \frac{\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-s}}{\sum_{\mathfrak{p}} N\mathfrak{p}^{-s}} = \delta,$$

where the denominator sum runs over all nonzero prime ideals of $\mathcal{O}_K$.

The denominator is asymptotically $\log\bigl(1/(s-1)\bigr)$ as $s \downarrow 1$ (Sharifi Prop. 7.1.12, p. 140); the lemmas below break out the analytic ingredients.

Lemma 1.1.2. *The higher-power tail of the log-Euler-product $\sum_{\mathfrak{p}, k \geq 2} \frac{N\mathfrak{p}^{-2s}}{1 - N\mathfrak{p}^{-s}}$ is bounded on a right neighbourhood of $s = 1$.*

Lemma 1.1.3. *For $s > 1$ in a right neighbourhood of 1,*

$$\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \;\leq\; [K : \mathbb{Q}] \sum_{p \text{ rational prime}} p^{-s}.$$

Proof. At most $[K : \mathbb{Q}]$ primes of $\mathcal{O}_K$ lie above any rational prime p (the identity $\sum_{\mathfrak{p}|p} e(\mathfrak{p} \mid p) f(\mathfrak{p} \mid p) = [K : \mathbb{Q}]$ in mathlib's `Ideal.sum_ramification_inertia`), and the norm of such a prime is at least p, so the bound follows by absolute convergence on $\mathrm{Re}(s) > 1$. ☐

Lemma 1.1.4. *As $s \downarrow 1$,*

$$\sum_{p \text{ rational prime}} p^{-s} \sim \log \frac{1}{s-1}.$$

Proof. The case $K = \mathbb{Q}$ of Lemma 1.1.10 below: combine the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ with the simple pole of ζ at $s = 1$. ☐

Lemma 1.1.5. *Euler-product-log identity: $\bigl|\log \zeta_K(s) - \sum_{\mathfrak{p}} N\mathfrak{p}^{-s}\bigr| \leq C$ on a right neighbourhood of 1.*

Proof. $\log \zeta_K(s) = \sum_{\mathfrak{p}} N\mathfrak{p}^{-s} + \sum_{\mathfrak{p}, k \geq 2} N\mathfrak{p}^{-ks}/k$; the tail is bounded by Lemma 1.1.2. ☐

Lemma 1.1.6. *Simple-pole identity: $\bigl|\log \zeta_K(s) - \log(1/(s-1))\bigr| \leq C$ on a right neighbourhood of 1.*

Proof. From the simple pole of ζ_K at $s = 1$ (mathlib's `NumberField.tendsto_sub_one_mul_dedekindZeta_nhdsGT`). ☐

Lemma 1.1.7. *There is a constant C such that, for $s > 1$ in a right neighbourhood of 1,*

$$\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \ \geq \ \log \frac{1}{s-1} - C.$$

Proof. Chain the two identities Lemmas 1.1.5 and 1.1.6: $\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \geq \log \zeta_K(s) - C_1 \geq \log(1/(s-1)) - C_1 - C_2$. ☐

Lemma 1.1.8. *There is a constant C' such that, for $s > 1$ in a right neighbourhood of 1,*

$$\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \ \leq \ \log \frac{1}{s-1} + C'.$$

Proof. Same two identities Lemmas 1.1.5 and 1.1.6, transposed. ☐

Lemma 1.1.9. *Let $f : (1, \infty) \to \mathbb{R}$. If f is sandwiched as $\log\bigl(1/(s-1)\bigr) - C \leq f(s) \leq \log\bigl(1/(s-1)\bigr) + C'$ on a right neighbourhood of 1, then $f(s)/\log\bigl(1/(s-1)\bigr) \to 1$ as $s \downarrow 1$.*

Proof. Since $\log\bigl(1/(s-1)\bigr) \to \infty$ as $s \downarrow 1$, the additive constants C, C' wash out under division, and the squeeze gives the limit 1. ☐

Lemma 1.1.10. *As $s \downarrow 1$,*

$$\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \sim \log \frac{1}{s-1}.$$

Proof. Combine the lower bound Lemma 1.1.7 and the upper bound Lemma 1.1.8 into a two-sided $\log\bigl(1/(s-1)\bigr) \pm O(1)$ sandwich, then apply Lemma 1.1.9 to extract the ratio limit. ☐

Density API

The following routine API lemmas are used by the proofs of the corollaries in Chapter 6.

Lemma 1.1.11. *A finite set of prime ideals has Dirichlet density 0.*

Proof. The numerator $\sum_{\mathfrak{p} \in S} \mathrm{N}(\mathfrak{p})^{-s}$ is bounded as $s \to 1^+$ (a finite sum of bounded terms), while the denominator $\sum_{\mathfrak{p}} \mathrm{N}(\mathfrak{p})^{-s} \to \infty$ by Lemma 1.1.10; the ratio tends to 0. ☐

Lemma 1.1.12. *If S has Dirichlet density δ, then its lower Dirichlet density is also δ.*

Proof. Convergence of the ratio implies the liminf and limsup coincide with the limit; specifically the liminf equals δ. ☐

Chapter 2

Decomposition, inertia, Frobenius

2.1 Decomposition group, inertia, Frobenius

Let L/K be a finite Galois extension of number fields with $G = \mathrm{Gal}(L/K)$. The Galois group acts on ideals of $\mathcal{O}_L$ (via the canonical G-action restricted to $\mathcal{O}_L$ and extended multiplicatively), and in particular permutes the primes lying above a given prime $\mathfrak{p}$ of $\mathcal{O}_K$.

Definition 2.1.1. *A nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$ is* unramified *in L if every prime $\mathfrak{P}$ of $\mathcal{O}_L$ lying over $\mathfrak{p}$ has ramification index 1, i.e. $\mathfrak{p}\mathcal{O}_L$ factors into distinct primes.*

Definition 2.1.2. *For a prime $\mathfrak{P}$ of $\mathcal{O}_L$, the* decomposition group *at $\mathfrak{P}$ is*

$$D_{\mathfrak{P}} \;=\; \{\sigma \in G : \sigma(\mathfrak{P}) = \mathfrak{P}\} \;=\; \mathrm{Stab}_G(\mathfrak{P}).$$

Definition 2.1.3. *The* inertia group *at $\mathfrak{P}$ is*

$$I_{\mathfrak{P}} \;=\; \{\sigma \in G \mid \forall x \in \mathcal{O}_L,\ \sigma(x) \equiv x \pmod{\mathfrak{P}}\}.$$

This is the generic ideal-inertia subgroup `Ideal.inertia` from mathlib, instantiated at the natural action of G on $\mathcal{O}_L$; acting trivially mod $\mathfrak{P}$ forces stabilising $\mathfrak{P}$, so $I_{\mathfrak{P}}$ sits inside $D_{\mathfrak{P}}$, and is the kernel of the action of $D_{\mathfrak{P}}$ on the residue field $\mathcal{O}_L/\mathfrak{P}$.

Lemma 2.1.4. *If $\mathfrak{P}$ is unramified over $\mathfrak{p} = \mathfrak{P} \cap \mathcal{O}_K$, then $I_{\mathfrak{P}} = \{1\}$.*

Proof. Mathlib's `Ideal.ramificationIdx_eq_one_of_isUnramifiedAt` gives the forward equivalence between unramifiedness and ramification index 1; combined with $|I_{\mathfrak{P}}| = e(\mathfrak{P} \mid \mathfrak{p})$ this yields the claim. □

When $\mathfrak{P}$ is unramified, the surjection $D_{\mathfrak{P}} \to \mathrm{Gal}((\mathcal{O}_L/\mathfrak{P})/(\mathcal{O}_K/\mathfrak{p}))$ (mathlib's `IsFractionRing.stabilizerHom_su` has trivial kernel Lemma 2.1.4, so it is an isomorphism. The residue Galois group of a finite-field extension is canonically cyclic, generated by the absolute Frobenius $x \mapsto x^{N\mathfrak{p}}$ (mathlib's `FiniteField.frobeniusAlgEquivO` Hence $D_{\mathfrak{P}}$ has a distinguished element.

Lemma 2.1.5. *Let $\mathfrak{P}$ be a nonzero prime of $\mathcal{O}_L$ with trivial inertia group $I_{\mathfrak{P}} = \{1\}$. There is a unique $\sigma \in D_{\mathfrak{P}}$ such that $\sigma(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$ for every $x \in \mathcal{O}_L$.*

Proof. With $I_{\mathfrak{P}}$ trivial, the surjection $D_{\mathfrak{P}} \to \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$ is an isomorphism, and the residue Galois group is cyclic with distinguished generator the absolute Frobenius; pull it back. □

Theorem 2.1.6. *Let $\mathfrak{P}$ be a nonzero prime of $\mathcal{O}_L$ lying over the unramified $\mathfrak{p}$. There is a unique $\sigma \in D_{\mathfrak{P}}$ such that*

$$\sigma(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}} \qquad \text{for every } x \in \mathcal{O}_L.$$

Proof. Unramifiedness gives $I_{\mathfrak{P}} = \{1\}$ (Lemma 2.1.4); apply Lemma 2.1.5. □

Definition 2.1.7. *For an unramified nonzero prime $\mathfrak{P}$ of $\mathcal{O}_L$, the* Frobenius automorphism $\mathrm{Frob}_{\mathfrak{P}} \in G$ *is the unique element of $D_{\mathfrak{P}}$ characterised by $\mathrm{Frob}_{\mathfrak{P}}(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_L$.*

Definition 2.1.8. *For a nonzero prime* $\mathfrak{p}$ *of* $\mathcal{O}_K$ *unramified in* L*, the* Frobenius conjugacy class *is*

$$\sigma_{\mathfrak{p}} \;=\; \{\mathrm{Frob}_{\mathfrak{P}} : \mathfrak{P} \mid \mathfrak{p}\} \;\in\; \mathrm{Conj}(G),$$

the conjugacy class in G *containing* $\mathrm{Frob}_{\mathfrak{P}}$ *for any (equivalently every) prime* $\mathfrak{P}$ *of* $\mathcal{O}_L$ *above* $\mathfrak{p}$*. The class is independent of the choice of* $\mathfrak{P}$*.*

Lemma 2.1.9. *For a nonzero prime* $\mathfrak{p}$ *of* $\mathcal{O}_K$ *unramified in* L*, there exists a conjugacy class* $C \subseteq \mathrm{Gal}(L/K)$ *such that for every prime* $\mathfrak{P}$ *of* $\mathcal{O}_L$ *above* $\mathfrak{p}$*,* $C = [\mathrm{Frob}_{\mathfrak{P}}]$*. (Equivalently, the Frobenius elements above* $\mathfrak{p}$ *are all conjugate.)*

Lemma 2.1.10. *For a nonzero prime* $\mathfrak{p}$ *of* $\mathcal{O}_K$ *unramified in* L*, and any prime* $\mathfrak{P}$ *of* $\mathcal{O}_L$ *above* $\mathfrak{p}$*,*

$$\sigma_{\mathfrak{p}} \;=\; [\mathrm{Frob}_{\mathfrak{P}}].$$

Proof. Unfold $\sigma_{\mathfrak{p}}$ at the unramified positive case to expose the choose from Lemma 2.1.9; the witness property of choose_spec at $\mathfrak{P}$ closes the goal. $\square$

Lemma 2.1.11. *Only finitely many nonzero primes of* $\mathcal{O}_K$ *ramify in* L*.*

Proof. The primes ramifying in L are exactly the divisors of the relative different ideal $\mathfrak{d}_{L/K} \subseteq \mathcal{O}_L$, which has a finite prime factorisation in the Dedekind domain $\mathcal{O}_L$. $\square$

Chapter 3

Zeta factorisation for abelian extensions

3.1 Zeta factorisation for an abelian extension

For an abelian Galois extension L/K of number fields with $G = \mathrm{Gal}(L/K)$, the Dedekind zeta function ζ_L factors as a product of Hecke L-functions indexed by characters of G. This is the analytic engine of the cyclotomic case of Chebotarev.

Definition 3.1.1. *A* character of G *is a group homomorphism* $\chi : G \to \mathbb{C}^\times$.

Lemma 3.1.2. *For an abelian character* $\chi : G \to \mathbb{C}^\times$*, there is a function* $L(\chi, \cdot) : \mathbb{C} \to \mathbb{C}$ *such that on* $\mathrm{Re}(s) > 1$,

$$L(\chi, s) \;=\; \prod_{\mathfrak{p}} \bigl(1 - \chi(\mathrm{Frob}_{\mathfrak{p}})\mathrm{N}(\mathfrak{p})^{-s}\bigr)^{-1},$$

where the product runs over primes $\mathfrak{p}$ *of* $\mathcal{O}_K$ *unramified in* L*, with the convention* $\chi(\mathfrak{p}) = 0$ *on ramified primes.*

Proof. Sharifi Prop. 7.1.18 (p. 141). The Euler product converges absolutely on $\mathrm{Re}(s) > 1$ by comparison with ζ_K. $\square$

Lemma 3.1.3. *For each unramified prime* $\mathfrak{p}$ *of* $\mathcal{O}_K$*, the local Euler factor of* ζ_L *at* $\mathfrak{p}$ *factors as*

$$\prod_{\mathfrak{P}|\mathfrak{p}} \bigl(1 - \mathrm{N}(\mathfrak{P})^{-s}\bigr)^{-1} \;=\; \prod_{\chi} \bigl(1 - \chi(\mathrm{Frob}_{\mathfrak{p}})\mathrm{N}(\mathfrak{p})^{-s}\bigr)^{-1}.$$

Proof. Standard cyclic-group character theory applied to the residue Galois group: the characters of $\mathrm{Gal}((\mathcal{O}_L/\mathfrak{P})/(\mathcal{O}_K/\mathfrak{p}))$ separate elements, and the indicated product over χ recovers the local factor (Sharifi 7.1.16 step at $\mathfrak{p}$). $\square$

Lemma 3.1.4. *For a nontrivial character* $\chi : G \to \mathbb{C}^\times$*, there is a constant* C *such that*

$$\Bigl\| \sum_{\mathrm{N}(\mathfrak{a}) \le N} \chi(\mathfrak{a}) \Bigr\| \;\le\; C \cdot N^{1-1/[K:\mathbb{Q}]} \qquad \textit{for all } N \ge 1.$$

Proof. Geometry-of-numbers count of ideals in a norm-bounded region (Sharifi 7.1.19, p. 142, step 1). Per-residue-class count $CN + O(N^{1-1/[K:\mathbb{Q}]})$; summing over $\chi(\mathfrak{a}) = \zeta$ for ζ a root of unity, the CN contributions cancel by $\sum_\zeta \zeta = 0$. $\square$

Lemma 3.1.5. *For every nontrivial character* $\chi : G \to \mathbb{C}^\times$*, the Dirichlet series* $L(\chi, s) = \sum_{\mathfrak{a}} \chi(\mathfrak{a})\mathrm{N}(\mathfrak{a})^{-s}$ *extends to a function analytic on* $\mathrm{Re}(s) > 1 - 1/[K : \mathbb{Q}]$*, agreeing with the Euler product on* $\mathrm{Re}(s) > 1$*. In particular,* $L(\chi, 1)$ *is well-defined.*

Proof. Combine Lemma 3.1.4 (Sharifi 7.1.19 step 1a) with Sharifi Lemma 7.1.5 (p. 138, a generic Dirichlet-series convergence criterion from a polynomial bound on partial sums): given $\left|\sum_{n\le N} a_n\right| \le CN^u$, the Dirichlet series $\sum_n a_n n^{-s}$ converges absolutely and uniformly on every compact subset of $\mathrm{Re}(s) > u$. Applied to $a_{\mathfrak{a}} = \chi(\mathfrak{a})$ with $u = 1 - 1/[K:\mathbb{Q}]$, the series defines an analytic function on $\mathrm{Re}(s) > 1 - 1/[K:\mathbb{Q}]$. □

Lemma 3.1.6. *For every nontrivial character $\chi : G \to \mathbb{C}^\times$, $L(\chi, 1) \neq 0$. (Here $L(\chi, 1)$ refers to the analytic extension of Lemma 3.1.5.)*

Proof. Sharifi 7.1.19 step 2 (p. 142). Write $\log \zeta_L(t) = \sum_\chi \log L(\chi, t)$ for $t > 1$. Up to a bounded function as $t \to 1^+$, the right side has absolute value at least $\left(1 - \sum_\chi m_\chi\right) \log(t-1)^{-1}$, where m_χ is the order of vanishing of $L(\chi, s)$ at $s = 1$. If any $m_\chi \ge 1$, this is at most $0 \cdot \log(t-1)^{-1}$, contradicting $\log \zeta_L(s) \sim \log(s-1)^{-1}$. □

Theorem 3.1.7. *Let L/K be a finite abelian Galois extension of number fields. There is a family of functions $\left\{L(\chi, \cdot) : \chi \in \widehat{G}\right\}$, each holomorphic on $\mathrm{Re}(s) > 1$, satisfying:*

1. $\zeta_L(s) = \displaystyle\prod_{\chi \in \widehat{G}} L(\chi, s)$ *for* $\mathrm{Re}(s) > 1$*;*
2. $L(\mathbf{1}, s) = \zeta_K(s)$*;*
3. $L(\chi, 1) \neq 0$ *for every* $\chi \neq \mathbf{1}$.

Proof. Compose Lemma 3.1.3 prime by prime, using the absolute convergence of the global Euler product; combine with the non-vanishing Lemma 3.1.6. □

Chapter 4

Chebotarev: cyclotomic case

4.1 Chebotarev: cyclotomic case

For a cyclotomic extension $L = K(\zeta_m)$, Chebotarev's theorem reduces directly to Dirichlet's argument. Source: Sharifi 7.2.1 (pp. 142–143).

Lemma 4.1.1. *Let $L = K(\zeta_m)$, $\mathfrak{p}$ a nonzero prime of $\mathcal{O}_K$ unramified in L, and $\mathfrak{P}$ a prime of $\mathcal{O}_L$ above $\mathfrak{p}$. Then for every primitive m-th root of unity $\zeta \in L$,*

$$\mathrm{Frob}_{\mathfrak{P}}(\zeta) \;=\; \zeta^{N\mathfrak{p}}.$$

Proof. By the defining property of $\mathrm{Frob}_{\mathfrak{P}}$ (acts as $N\mathfrak{p}$-power on $\mathcal{O}_L/\mathfrak{P}$) combined with the fact that ζ_m lifts isomorphically from the residue field for $\mathfrak{p} \nmid m$. □

Lemma 4.1.2. *For every character χ of G, the partial sum $\sum_{\mathfrak{p}} \chi(\mathrm{Frob}_{\mathfrak{p}}) N\mathfrak{p}^{-s}$ over $\mathfrak{p}$ unramified in L is bounded by $C \log(1/(s-1)) + C$ for some constant C, in a right neighbourhood of $s = 1$.*

Proof. Expand $\log L(\chi, s)$ via the Euler product Lemma 3.1.2; the higher-power tail is bounded analogously to Lemma 1.1.2. □

Lemma 4.1.3. *Let $L = K(\zeta_m)$, $\sigma \in G$, and $\mathfrak{p}$ a nonzero prime of $\mathcal{O}_K$ unramified in L with $\sigma_{\mathfrak{p}}$ the conjugacy class of σ. Then*

$$\sum_{\chi \in \widehat{G}} \chi(\sigma)\chi(\mathrm{Frob}_{\mathfrak{p}})^{-1} \;=\; |G|\,.$$

Lemma 4.1.4. *Let $L = K(\zeta_m)$, $\sigma \in G$, and $\mathfrak{p}$ a nonzero prime of $\mathcal{O}_K$ unramified in L whose Frobenius class is not the conjugacy class of σ. Then*

$$\sum_{\chi \in \widehat{G}} \chi(\sigma)\chi(\mathrm{Frob}_{\mathfrak{p}})^{-1} \;=\; 0.$$

Proof. Standard finite-group character orthogonality for the dual of G (Sharifi 7.2.1 step at p. 142). Under the isomorphism $G \cong (\mathbb{Z}/m\mathbb{Z})^{\times}$ from Lemma 4.1.1, $\chi(\mathrm{Frob}_{\mathfrak{p}})$ depends only on $N\mathfrak{p} \bmod m$, and the sum reduces to the orthogonality relation on the cyclic group $(\mathbb{Z}/m\mathbb{Z})^{\times}$. □

Lemma 4.1.5. *Let $L = K(\zeta_m)$ and $\sigma \in G$. The Frobenius-fibre prime sum is asymptotic to $(1/|G|)\log(1/(s-1))$:*

$$\lim_{s \downarrow 1} \frac{\sum_{\mathfrak{p}:\, \sigma_{\mathfrak{p}} = \sigma} N\mathfrak{p}^{-s}}{\log(1/(s-1))} \;=\; \frac{1}{|G|}.$$

Proof. Multiply $\log L(\chi, s)$ by $\chi(\sigma)^{-1}$ and sum over χ: the orthogonality relations (Lemmas 4.1.3 and 4.1.4) pick out the primes with Frobenius σ, giving $\sum_{\chi} \chi(\sigma)^{-1} \log L(\chi, s) \sim |G| \sum_{\sigma_{\mathfrak{p}} = \sigma} N\mathfrak{p}^{-s}$; on the other hand it $\sim \log \zeta_K(s) \sim \log(1/(s-1))$ (only $\chi = 1$ contributes a pole, by Lemma 3.1.6). Divide. □

Lemma 4.1.6. *If* $\mathrm{num}(s)/\log(1/(s-1)) \to c$ *and* $\mathrm{den}(s)/\log(1/(s-1)) \to 1$ *as* $s \downarrow 1$*, then* $\mathrm{num}(s)/\mathrm{den}(s) \to c$.

Proof. Write $\mathrm{num}/\mathrm{den} = (\mathrm{num}/L)/(\mathrm{den}/L)$ with $L = \log(1/(s-1))$, which is positive (hence nonzero) for $s \in (1,2)$ since $1/(s-1) > 1$; apply the quotient rule for limits and cancel L. ☐

Lemma 4.1.7. *Let* $L = K(\zeta_m)$ *and* $\sigma \in G$*. Then*

$$\lim_{s\downarrow 1} \frac{\sum_{\mathfrak{p}:\, \sigma_{\mathfrak{p}}=\sigma} N\mathfrak{p}^{-s}}{\sum_{\mathfrak{p}} N\mathfrak{p}^{-s}} \;=\; \frac{1}{|G|}.$$

Proof. The numerator asymptotic Lemma 4.1.5 ($\mathrm{num} \sim (1/|G|)\log(1/(s-1))$) and the denominator asymptotic Lemma 1.1.10 ($\mathrm{den} \sim \log(1/(s-1))$) feed the ratio glue Lemma 4.1.6. ☐

Theorem 4.1.8. *For* K *a number field,* $m \geq 1$*,* $L = K(\zeta_m)$*, and every* $\sigma \in G = \mathrm{Gal}(L/K)$*,*

$$\delta\big(\{\mathfrak{p} \subset \mathcal{O}_K \mid \sigma_{\mathfrak{p}} = \sigma\}\big) \;=\; \frac{1}{|G|}.$$

Chapter 5

Chebotarev: abelian case

5.1 Chebotarev: abelian case via cyclotomic crossing

The abelian case is reduced to the cyclotomic case by Chebotarev's original crossing trick. Source: Sharifi 7.2.2 Step 2 (pp. 143–144).

Lemma 5.1.1. *Let G, H be finite groups, $\sigma \in G$, $\tau \in H$. If $|G| \mid \operatorname{ord}(\tau)$, then $\langle(\sigma,\tau)\rangle \cap (G \times \{1\}) = \{1\}$.*

Proof. Pure group theory. If $(\sigma^k, \tau^k) \in G \times \{1\}$ then $\tau^k = 1$, so $\operatorname{ord}(\tau) \mid k$, hence $|G| \mid k$, hence $\sigma^k = 1$. □

Lemma 5.1.2. *Let L/K be a finite abelian Galois extension with $G = \operatorname{Gal}(L/K)$, $\sigma \in G$, and $m \geq 1$ coprime to the discriminant of L, so that $\operatorname{Gal}(L(\zeta_m)/K) \cong G \times H$ with $H = \operatorname{Gal}(K(\zeta_m)/K) \subseteq (\mathbb{Z}/m\mathbb{Z})^\times$. Set $H_n = \{\tau \in H \mid |G| \mid \operatorname{ord}(\tau)\}$. Then*

$$\delta_{\inf}\big(\{\mathfrak{p} \subset \mathcal{O}_K \mid \sigma_\mathfrak{p} = \sigma\}\big) \;\geq\; \frac{|H_n|}{|G| \cdot |H|}.$$

Corrected. *An earlier draft of this lemma claimed $\delta\big(\{\mathfrak{p} : \sigma_\mathfrak{p} = \sigma\}\big) = 1/(|G| \cdot |H|)$ — that is mathematically wrong (the set $\{\sigma_\mathfrak{p} = \sigma\}$ has density $1/|G|$, not $1/(|G| \cdot |H|)$). The actual per-m step that feeds into the proof of $\delta(\sigma_\mathfrak{p} = \sigma) = 1/|G|$ is the* $\liminf$ *lower bound above (Sharifi p. 144).*

Proof. By Lemma 5.1.1, the fixed field $F = L(\zeta_m)^{\langle(\sigma,\tau)\rangle}$ satisfies $F(\zeta_m) = L(\zeta_m)$, so the extension $L(\zeta_m)/F$ is cyclotomic. Apply the cyclotomic case Theorem 4.1.8 to $L(\zeta_m)/F$ to obtain $\delta_F = 1/|\langle(\sigma,\tau)\rangle|$. The conjugacy-class reduction (Theorem 6.1.3 below, Step 1's counting) lifts this to a K-density of $1/(|G| \cdot |H|)$. □

Lemma 5.1.3. *Let $n = p_1^{k_1} \cdots p_r^{k_r}$ with p_i distinct primes, $k_i \geq 1$. For an integer $m \geq 1$ with $m \equiv 1 \pmod{n^j}$, setting $j_i = v_{p_i}(m-1) \geq j$ and $H_n = \{\tau \in (\mathbb{Z}/m\mathbb{Z})^\times \mid n \mid \operatorname{ord}(\tau)\}$,*

$$\frac{|H_n|}{|(\mathbb{Z}/m\mathbb{Z})^\times|} \;=\; \prod_{i=1}^{r}\left(1 - \frac{p_i^{k_i-1}}{p_i^{j_i k_i}}\right) \;\geq\; \prod_{i=1}^{r}\left(1 - \frac{1}{p_i^{(j-1)k_i+1}}\right).$$

Proof. Direct combinatorial computation on $(\mathbb{Z}/m\mathbb{Z})^\times$ using CRT and the prime-power factorisation of n (Sharifi p. 144). □

Lemma 5.1.4. *As $k \to \infty$, $|H_n| / \big|(\mathbb{Z}/n^k\mathbb{Z})^\times\big| \to 1$.*

Proof. Limit of the product formula Lemma 5.1.3 as $j \to \infty$: each factor tends to 1. □

Lemma 5.1.5. *For every $\sigma \in G$, $\delta_{\inf}\big(\{\mathfrak{p} \mid \sigma_\mathfrak{p} = \sigma\}\big) \geq 1/|G|$.*

Proof. Take the limit of the per-m bound Lemma 5.1.2 as $m \to \infty$ along $m \equiv 1 \pmod{|G|^k}$, where $|H_n| / |H| \to 1$ by Lemma 5.1.4. □

Lemma 5.1.6. *As $s \downarrow 1$, the sum over $\sigma \in G$ of the density ratios of the fibres $\{\mathfrak{p} \mid \sigma_{\mathfrak{p}} = \sigma\}$ tends to 1.*

Proof. The fibres partition the unramified primes; the ramified primes are finite (Lemma 2.1.11) hence contribute density 0, so the partial sums of the fibre ratios equal the ratio for the unramified primes, which $\to 1$. $\square$

Lemma 5.1.7. *Let g_i (i in a finite index set of size N) be real functions with $\liminf_{s\downarrow 1} g_i \geq 1/N$ for each i, and $\sum_i g_i(s) \to 1$ as $s \downarrow 1$. Then $g_i(s) \to 1/N$ for every i.*

Proof. Pure real analysis: the lower bounds and the sum-limit pin every g_i to $1/N$ by a $\liminf/\limsup$ pigeonhole. $\square$

Theorem 5.1.8. *For a finite abelian Galois extension L/K of number fields with $G = \mathrm{Gal}(L/K)$ and every $\sigma \in G$,*

$$\delta\big(\{\mathfrak{p} \subset \mathcal{O}_K \mid \sigma_{\mathfrak{p}} = \sigma\}\big) \;=\; \frac{1}{|G|}.$$

Proof. The $|G|$ fibres each have $\liminf \geq 1/|G|$ (Lemma 5.1.5) and their density ratios sum to 1 (Lemma 5.1.6); the pigeonhole glue Lemma 5.1.7 forces each fibre's ratio to the limit $1/|G|$. $\square$

Chapter 6

Chebotarev density theorem

6.1 Chebotarev density theorem

The general Galois case is reduced to the abelian case via the cyclic subextension generated by a class representative. Source: Sharifi 7.2.2 Step 1 (p. 143).

Lemma 6.1.1. *Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$, $C \subseteq G$ a conjugacy class, $\sigma \in C$, and $\mathfrak{p}$ a nonzero prime of $\mathcal{O}_K$ unramified in L with $\sigma_{\mathfrak{p}} = C$. The number of primes $\mathfrak{P}$ of $\mathcal{O}_L$ above $\mathfrak{p}$ with $\mathrm{Frob}_{\mathfrak{P}} = \sigma$ is exactly $|G| / (f \cdot |C|)$, where $f = \mathrm{ord}(\sigma)$.*

Proof. The Galois group G acts transitively on primes above $\mathfrak{p}$, with stabiliser $D_{\mathfrak{P}}$. Hence the orbit has size $|G| / |D_{\mathfrak{P}}|$. For unramified $\mathfrak{p}$, $D_{\mathfrak{P}} = \langle \mathrm{Frob}_{\mathfrak{P}} \rangle$ has order f, so there are $|G| / f$ primes above $\mathfrak{p}$. Their Frobenius elements form a G-orbit in C (under conjugation), which is all of C by transitivity; this orbit has size $|C|$, and within each orbit the count of representatives σ is $|G| / (f \cdot |C|)$ (Sharifi p. 143). ☐

Lemma 6.1.2 (Density lift through the fixed field)**.** *Let $\sigma \in \mathrm{Gal}(L/K)$, $E = L^{\langle \sigma \rangle}$, and $\sigma_E \in \mathrm{Gal}(L/E)$ the corresponding element. Given the abelian-case density over E for the Frobenius-fibre of σ_E (value $1/|\mathrm{Gal}(L/E)|$), the K-density of the Frobenius class $[\sigma]$ is $|[\sigma]| / |G|$.*

Proof. For primes P of E above $\mathfrak{p} \in S = \{\mathfrak{p} \mid \sigma_{\mathfrak{p}} = [\sigma]\}$, exactly $|G| / (f\,|C|)$ of the primes of L above have Frobenius σ (Lemma 6.1.1); with $\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \sim \sum_P NP^{-s}$ (Lemma 1.1.10 over both K and E), $\delta_K(S) = (f\,|C| / |G|)\,\delta_E(T_\sigma) = (f\,|C| / |G|)(1/f) = |C| / |G|$. ☐

Theorem 6.1.3 (Chebotarev density theorem)**.** *For a finite Galois extension L/K of number fields with $G = \mathrm{Gal}(L/K)$ and any conjugacy class $C \subseteq G$,*

$$\delta\big(\{\mathfrak{p} \subset \mathcal{O}_K \mid \sigma_{\mathfrak{p}} = C\}\big) \;=\; \frac{|C|}{|G|}.$$

Proof. Pick a representative $\sigma \in C$ (`ConjClasses.mk_surjective`) and let $E = L^{\langle \sigma \rangle}$ be the fixed field of $\langle \sigma \rangle$. The map $\langle \sigma \rangle \xrightarrow{\sim} \mathrm{Gal}(L/E)$ (`IntermediateField.subgroupEquivAlgEquiv`) shows $\mathrm{Gal}(L/E)$ is commutative (as $\langle \sigma \rangle$ is cyclic), so L/E is abelian and Theorem 5.1.8 applies to its Frobenius σ_E, giving $\delta_E(T_\sigma) = 1/|\mathrm{Gal}(L/E)|$. Feeding this into the density lift Lemma 6.1.2 yields $|C| / |G|$. ☐

Theorem 6.1.4 (Chebotarev, abelian case)**.** *When $G = \mathrm{Gal}(L/K)$ is abelian, the Chebotarev density theorem holds: for every conjugacy class C, $\delta(\{\mathfrak{p} \mid \sigma_{\mathfrak{p}} = C\}) = |C| / |G|$.*

Proof. In an abelian group each class $C = [\sigma]$ is a singleton (Lemma 6.1.6), so $|C| = 1$ and the target density is $1/|G|$; this is exactly Theorem 5.1.8 for the representative σ. ☐

Corollaries

The corollaries below are derived directly by composition from Theorem 6.1.3. The Lean derivations use four small helper sub-lemmas captured here.

Lemma 6.1.5. *Let G be a finite group. The conjugacy class of the identity has cardinality 1: $\left|(\mathrm{Conj}(G))_{\{1\}}\right| = 1$.*

Proof. The carrier is $\{1\}$ (by `isConj_one_left`, a conjugate to 1 iff $a = 1$); a singleton has cardinality 1. ☐

Lemma 6.1.6. *In a commutative finite group, every conjugacy class is a singleton, so has cardinality 1.*

Proof. By `isConj_iff_eq` (conjugacy is equality in a commutative monoid), the carrier of $[g]$ is $\{g\}$. ☐

Lemma 6.1.7. *For a finite-dimensional Galois extension L/K, $|\mathrm{Gal}(L/K)| = [L : K]$.*

Proof. Mathlib's `IsGalois.card_aut_eq_finrank`. ☐

Lemma 6.1.8. *A set S of prime ideals with positive Dirichlet density is infinite.*

Proof. Contrapositive: a finite set of primes has density 0 by Lemma 1.1.11 (the finite case of Definition 1.1.1). ☐

Lemma 6.1.9. *Every conjugacy class in a finite group has cardinality at least 1 (it contains at least its own representative).*

Proof. A representative a with $[a] = C$ lies in the carrier (`ConjClasses.mk_surjective` + `mem_carrier_iff_mk_eq`), so the carrier is nonempty; a nonempty finite set has positive cardinality. ☐

Corollary 6.1.10. *For a finite Galois extension L/K of number fields, the Dirichlet density of primes of $\mathcal{O}_K$ that split completely in L equals $1/[L : K]$.*

Proof. A prime $\mathfrak{p}$ splits completely in L iff every $\mathfrak{P} \mid \mathfrak{p}$ has $\mathrm{Frob}_{\mathfrak{P}} = 1$, i.e. the Frobenius conjugacy class is the trivial class. Apply Theorem 6.1.3 at $C = \mathrm{Conj}(G)_{\{1\}}$; by Lemma 6.1.5 the numerator $|C| = 1$ and by Lemma 6.1.7 the denominator $|G| = [L : K]$, so the density equals $1/[L : K]$. ☐

Corollary 6.1.11. *For a finite Galois extension L/K of number fields and any conjugacy class $C \subseteq \mathrm{Gal}(L/K)$, the set of primes $\mathfrak{p}$ of $\mathcal{O}_K$ unramified in L with Frobenius conjugacy class C is infinite.*

Proof. Theorem 6.1.3 gives Dirichlet density $|C|\,/\,|G|$, which is positive since $|C| \geq 1$ (Lemma 6.1.9) and $|G| \geq 1$. Apply Lemma 6.1.8. ☐

Corollary 6.1.12 (Dirichlet, density form)**.** *For $n \geq 1$ and $a \in (\mathbb{Z}/n\mathbb{Z})^\times$, the Dirichlet density of the rational primes p with $p \equiv a \pmod{n}$ equals $1/\varphi(n)$.*

Proof. Specialise Theorem 6.1.3 to $K = \mathbb{Q}$, $L = \mathbb{Q}(\zeta_n)$. The cyclotomic character identifies $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ with $(\mathbb{Z}/n\mathbb{Z})^\times$; under this identification, the Frobenius σ_p at unramified p corresponds to $p \bmod n$. The Galois group is abelian, so each conjugacy class is a singleton; the set of primes with $p \equiv a \pmod{n}$ is the Frobenius preimage of $\{a\}$, whose density is $1/\varphi(n)$. ☐